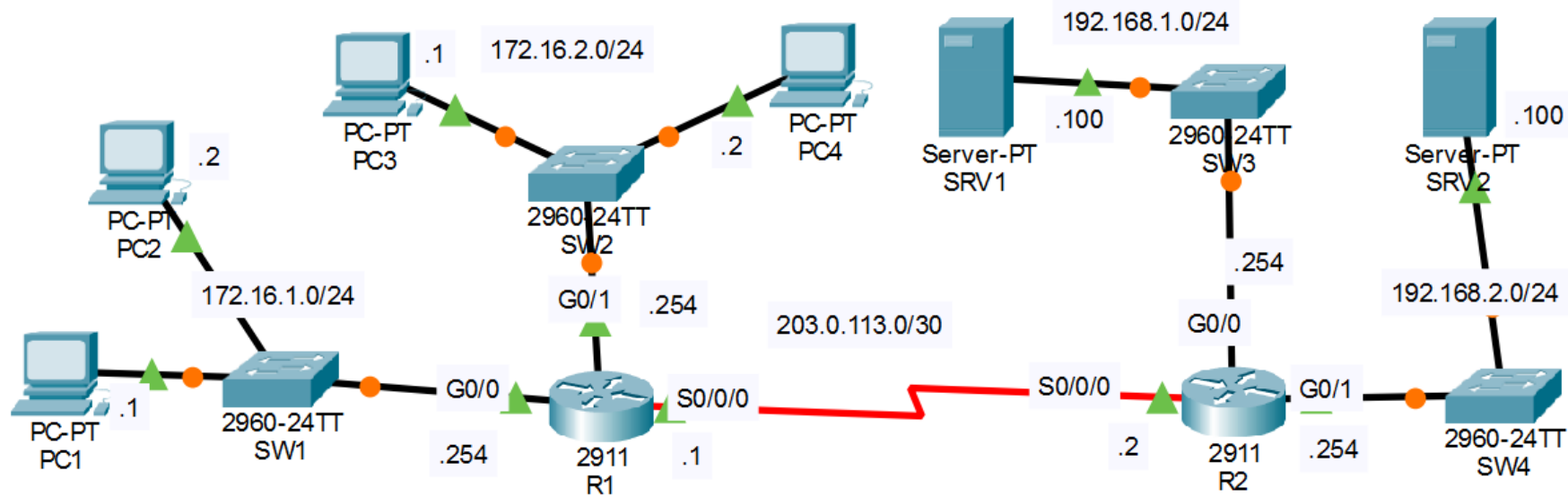
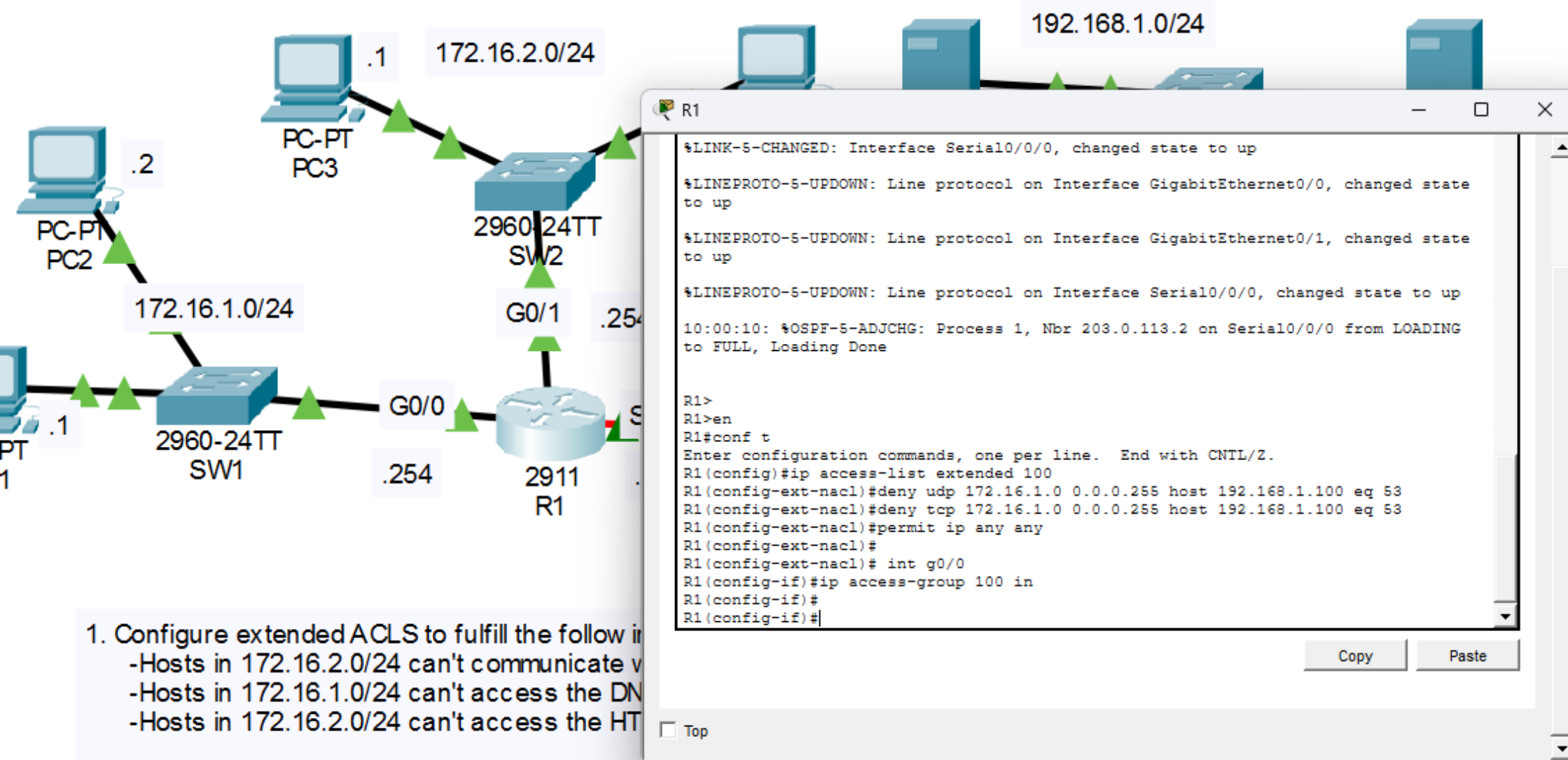


Network Topology



1. Configure extended ACLS to fulfill the following network policies:
 - Hosts in 172.16.2.0/24 can't communicate with PC1.
 - Hosts in 172.16.1.0/24 can't access the DNS service on SRV1.
 - Hosts in 172.16.2.0/24 can't access the HTTP or HTTPS services on SRV2.

Created a ACL to deny access from Hosts in 172.16.1.0/24 to Server 1





```
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>
```

Verify is DNS is blocked

```
C:\>ping 192.168.1.100
```

```
Pinging 192.168.1.100 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 192.168.1.100: bytes=32 time=7ms TTL=126
```

```
Reply from 192.168.1.100: bytes=32 time=9ms TTL=126
```

```
Reply from 192.168.1.100: bytes=32 time=8ms TTL=126
```

```
Ping statistics for 192.168.1.100:
```

```
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 7ms, Maximum = 9ms, Average = 8ms
```

```
C:\>ping srv1
```

```
Ping request could not find host srv1. Please check the name and try again.
```

```
C:\>
```

IP ping works

DNS Is blocked





R1



```
R1(config-ext-nacl)#deny tcp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq 53
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#
R1(config-ext-nacl)# int g0/0
R1(config-if)#ip access-group 100 in
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#ip access-list extended 101
R1(config-ext-nacl)#deny ip 172.16.2.0 0.0.0.255 host 172.16.1.1
R1(config-ext-nacl)#deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq 80
R1(config-ext-nacl)#deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq 443
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#
R1(config-ext-nacl)#int g0/1
R1(config-if)#ip access-group 101 in
R1(config-if)#
```

Configured and applied Extended ACL 101 on R1 to block traffic from 172.16.2.0/24 to PC1 and HTTP/HTTPS access to SRV2 while allowing other traffic.

Copy

Paste



Top



PC3



Physical

Config

Desktop

Programming

Attributes

Web Browser

X

<

>

URL

http://Cisco.com

Go

Stop

Request Timeout

Verify is HTTP/HTTPS is blocked



Physical

Config

Desktop

Programming

Attributes

Command Prompt

X

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.2.254: Destination host unreachable.

Reply from 172.16.2.254: Destination host unreachable.

Reply from 172.16.2.254: Destination host unreachable.

Reply from 172.16.2.254: Destination host unreachable.

Ping statistics for 172.16.1.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

Verify is PC3 can ping/communicate with PC1

```
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#ip access-list extended 101
R1(config-ext-nacl)#deny ip 172.16.2.0 0.0.0.255 host 172.16.1.1
R1(config-ext-nacl)#deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq 80
R1(config-ext-nacl)#deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq 443
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#
R1(config-ext-nacl)#int g0/1
R1(config-if)#ip access-group 101 in
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#
R1(config-if)#do sh access-list
Extended IP access list 100
  10 deny udp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq domain (12 match(es))
  20 deny tcp 172.16.1.0 0.0.0.255 host 192.168.1.100 eq domain
  30 permit ip any any (4 match(es))
Extended IP access list 101
  10 deny ip 172.16.2.0 0.0.0.255 host 172.16.1.1 (4 match(es))
  20 deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq www (80 match(es))
  30 deny tcp 172.16.2.0 0.0.0.255 host 192.168.2.100 eq 443
  40 permit ip any any (6 match(es))

R1(config-if)#
```

Verified the configured Extended ACLs on R1 and their traffic restrictions.